Assignment 4: Kalman Filter and Parameter Estimation

02417 Time Series Analysis — Spring 2026
Technical University of Denmark

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1 Parameter estimation in a 1-D state-space model

We study the scalar linear Gaussian state-space model

$$\begin{aligned} X_t &= aX_{t-1} + b + e_{1,t}, & e_{1,t} &\sim \mathcal{N}(0, \sigma_1^2), \\ Y_t &= X_t + e_{2,t}, & e_{2,t} &\sim \mathcal{N}(0, \sigma_2^2 = 1). \end{aligned} \tag{1}$$

1.1 Simulating the latent process (1.1)

With $a = 0.9$, $b = 1$, $\sigma_1 = 1$ and $X_0 = 5$, we simulate five independent trajectories over $n = 100$ steps (Fig. 1). All paths relax around the stationary mean $b/(1-a) = 10$; short-term fluctuations differ because of different noise realisations, but the long-run level and the mean-reversion rate are the same.

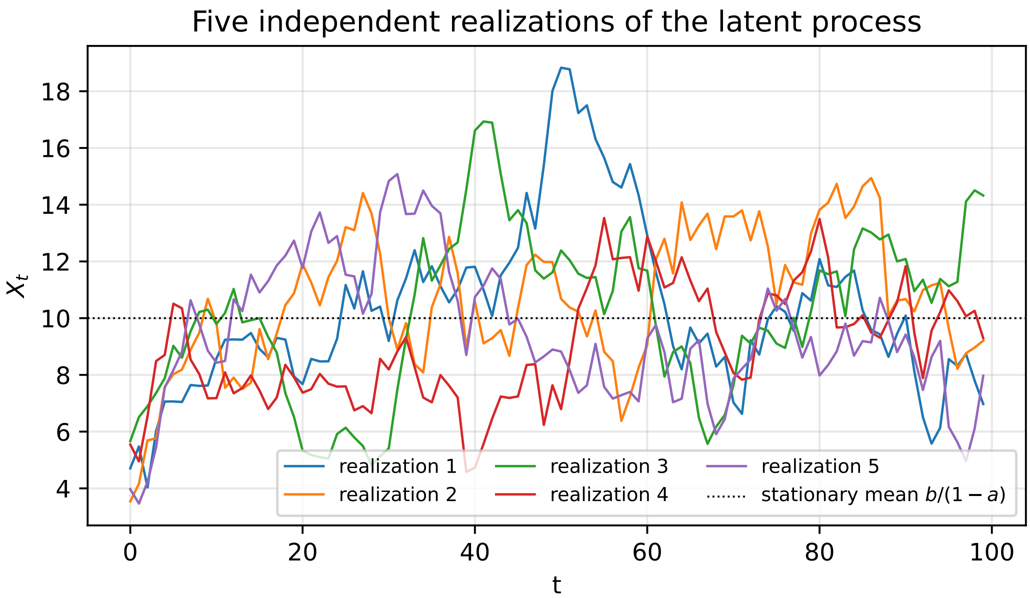


Figure 1: Five independent realizations of (1) with $a = 0.9, b = 1, \sigma_1 = 1$.

1.2 Adding observation noise (1.2)

Adding independent $\mathcal{N}(0, 1)$ measurement noise yields Fig. 2. The observations scatter around the latent trajectory; ordinary least-squares on Y_t would be *biased* because the regressor is measured with error. This motivates the Kalman filter.

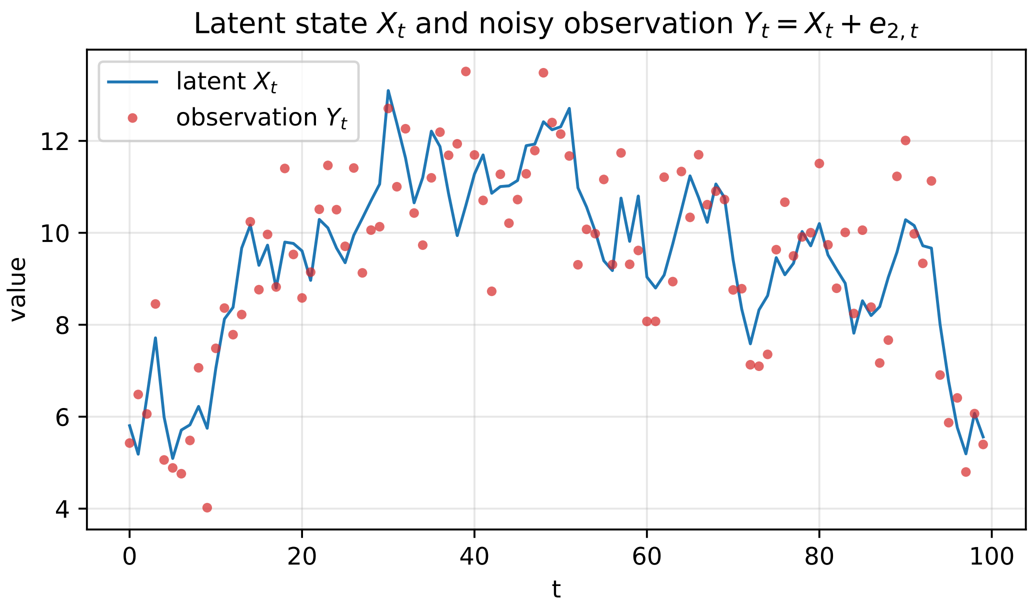


Figure 2: Latent state and noisy observation (single realization).

1.3 Kalman filter (1.3)

For (1) with $C = 1$ the scalar recursions are

$$\begin{aligned}
\text{Predict:} \quad & \hat{X}_{t|t-1} = a\hat{X}_{t-1|t-1} + b, \quad P_{t|t-1} = a^2 P_{t-1|t-1} + \sigma_1^2, \\
\text{Innovate:} \quad & \epsilon_t = Y_t - \hat{X}_{t|t-1}, \qquad S_t = P_{t|t-1} + \sigma_2^2, \\
\text{Update:} \quad & K_t = P_{t|t-1}/S_t, \qquad \hat{X}_{t|t} = \hat{X}_{t|t-1} + K_t\epsilon_t, \quad P_{t|t} = (1 - K_t)P_{t|t-1}.
\end{aligned}$$

Applied to the series from §1.2 (true parameters, $\hat{X}_0 = 5$, $P_0 = 10$) the filter tracks the latent state closely (Fig. 3). The 95% prediction band contains almost every observation; the standardized innovations pass the Ljung–Box test at lag 10 ($Q = 10.0$, $p = 0.44$), confirming that the filter is correctly specified.

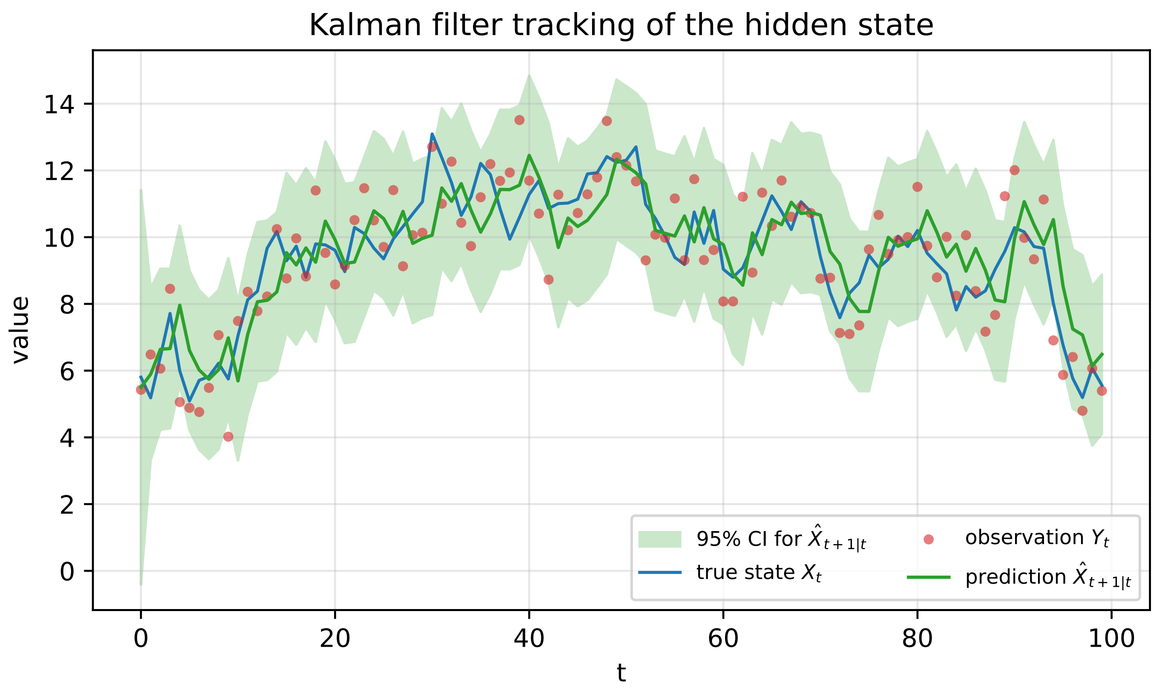


Figure 3: One-step predictions, 95% CI, true state and observations.

1.4 Maximum-likelihood estimation (1.4)

Using the prediction-error decomposition the log-likelihood is

$$\log L(\theta) = -\tfrac{1}{2} \sum_{t=1}^{n} \Big[\log(2\pi S_t) + \epsilon_t^2 / S_t \Big],$$

with $\theta = (a, b, \sigma_1)$. We minimize $-\log L$ with a two-stage routine (`Nelder-Mead` $\to$ `L-BFGS-B`); σ_1 is parameterised as $\exp(\log \sigma_1)$ to enforce positivity. For each of three parameter settings we simulate 100 independent series of length $n = 100$ and estimate the parameters; results are summarised in Fig. 4 and Table 1.

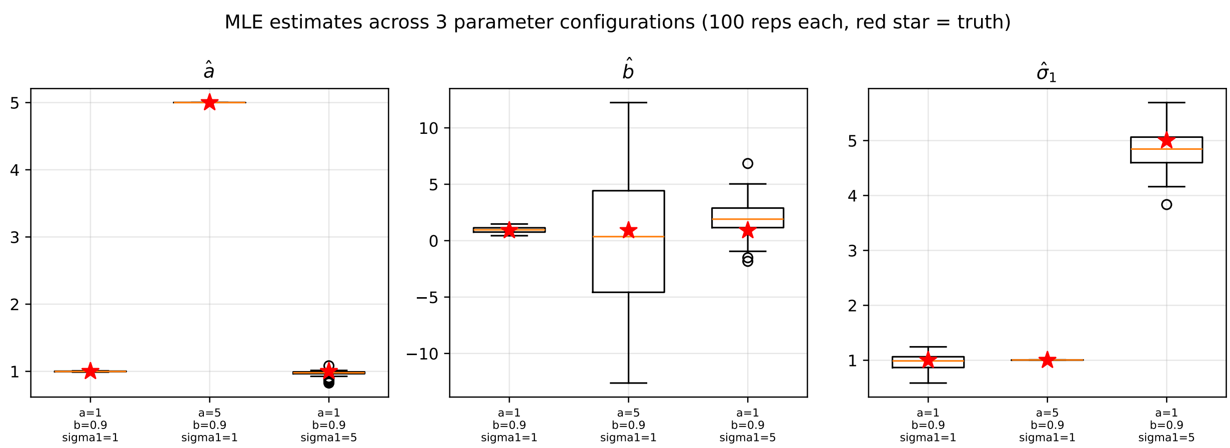


Figure 4: Boxplots of estimated $\hat{a}, \hat{b}, \hat{\sigma}_1$ across three cases (red star = truth).

Table 1: Monte-Carlo summary (100 reps, $n = 100$).

Case	Parameter	True	Mean	Bias	SD	RMSE
a=1, b=0.9, σ_1=1	a	1.000	0.999	-0.001	0.004	0.004
	b	0.900	0.955	+0.055	0.233	0.239
	σ_1	1.000	0.960	-0.040	0.148	0.153
a=5, b=0.9, σ_1=1	a	5.000	5.000	+0.000	0.000	0.000
	b	0.900	0.150	-0.750	5.747	5.767
	σ_1	1.000	1.006	+0.006	0.002	0.006
a=1, b=0.9, σ_1=5	a	1.000	0.972	-0.028	0.037	0.046
	b	0.900	2.062	+1.162	1.431	1.838
	σ_1	5.000	4.830	-0.170	0.343	0.381

Discussion. Case 1 ($a = 1, b = 0.9, \sigma_1 = 1$) sits on the stationarity boundary. The process then behaves like a random walk with drift, so a and b are only weakly identifiable — both have wide dispersion yet small mean bias. Case 2 ($a = 5$) is explosive: X_t grows geometrically, so the signal swamps the noise and $\hat{a}$ concentrates essentially at the true value to machine precision (SD $\approx 10^{-16}$). However, as $a > 1$ the drift b is dwarfed by the dominant aX_{t-1} term and the Monte-Carlo spread of $\hat{b}$ is huge (SD ≈ 5.7); we use a different random initial value of b for the optimizer at every replication precisely to expose this likelihood ridge. Case 3 ($\sigma_1 = 5$) has large process noise relative to the observation noise; σ_1 is easy to recover but a is harder because persistence is masked by dominant shocks. Overall σ_1 is the most consistently estimated parameter; a and b are correlated via the stationary mean $b/(1-a)$, which makes them jointly only weakly identified whenever the process is near-unit-root or explosive.

1.5 Robustness to heavy-tailed process noise (1.5)

Replacing $e_{1,t}$ by a scaled Student-t variable, $X_t = aX_{t-1} + b + \sigma_1\lambda_t$ with $\lambda_t \sim t_\nu$, keeps the observation model Gaussian. Figure 5 compares the densities: tails thicken rapidly as ν decreases; for $\nu = 1$ (Cauchy) the density has infinite variance.

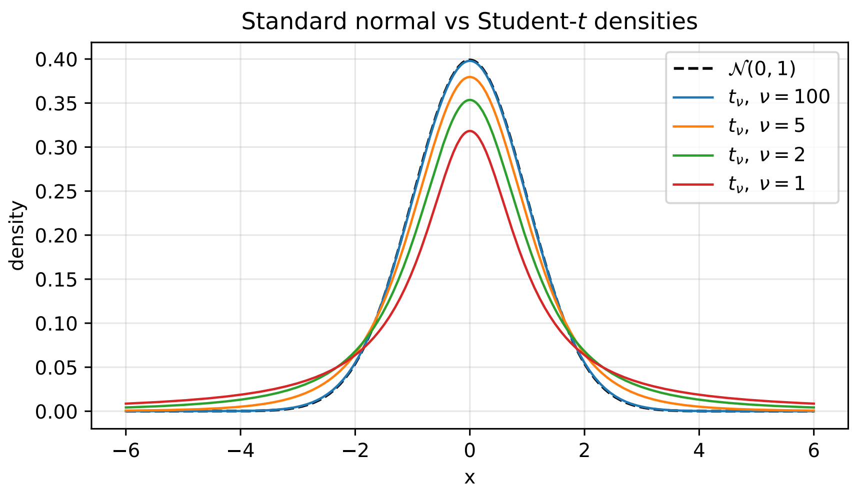


Figure 5: Student-t densities vs. standard normal.

We re-run the estimation pipeline assuming (incorrectly) Gaussian process noise (Fig. 6, Table 2).

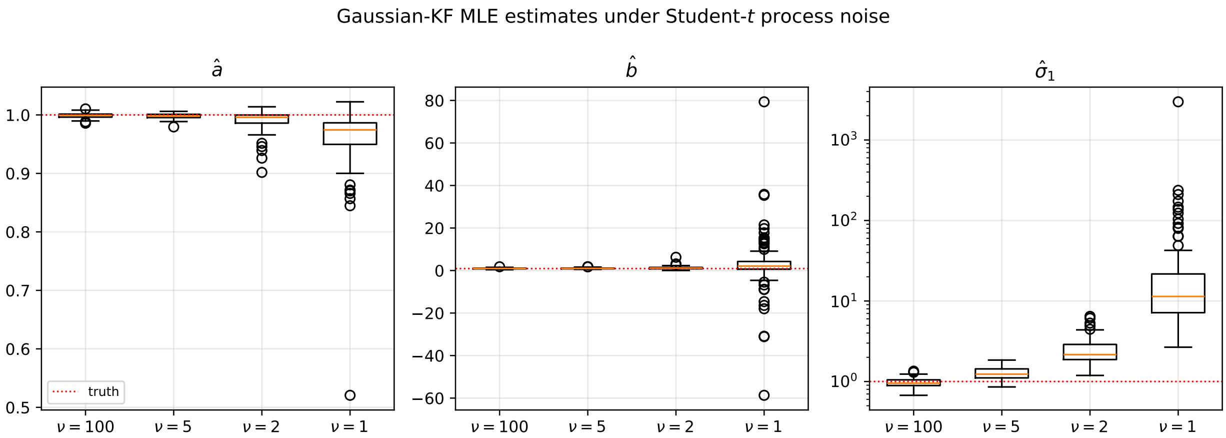


Figure 6: Estimates under Student-t process noise (log-scale for $\hat{\sigma}_1$).

Table 2: Monte-Carlo summary under Student-t noise, $n = 100$, 100 reps.

ν	Parameter	Mean	Bias	Success rate
100	a	0.999	-0.001	100%
	b	0.959	+0.059	100%
	σ_1	0.966	-0.034	100%
5	a	0.998	-0.002	100%
	b	1.019	+0.119	100%
	σ_1	1.270	+0.270	100%
2	a	0.990	-0.010	100%
	b	1.265	+0.365	100%
	σ_1	2.531	+1.531	100%
1	a	0.960	-0.040	100%
	b	2.478	+1.578	100%
	σ_1	55.989	+54.989	100%

Discussion. $\hat{a}$ remains nearly unbiased, but $\hat{\sigma}_1$ is progressively *inflated* as the tails become heavier: the Gaussian likelihood attributes occasional large innovations to a larger system-noise variance rather than to heavy tails. For $\nu = 1$ (Cauchy) the population variance does not exist, so $\hat{\sigma}_1$ diverges (median around 56). In practice: the Kalman filter still tracks the mean well, but uncertainty estimates and parameter inference become unreliable if residuals show heavy tails — a robust observation model (e.g. Student-t filter, M-estimation, outlier rejection) would be preferable.

2 Transformer station temperature

2.1 Exploratory analysis (2.1)

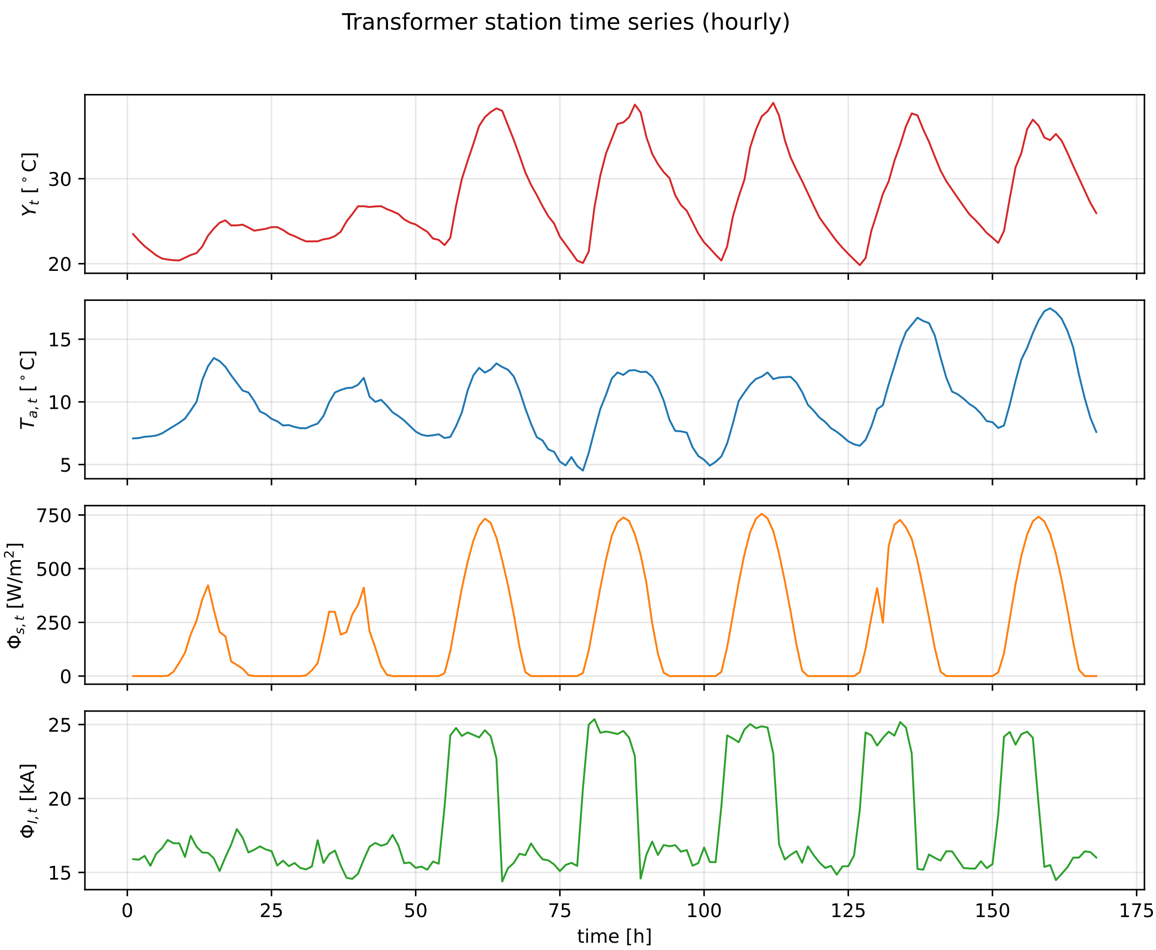


Figure 7: Transformer temperature Y_t, outdoor temperature $T_{a,t}$, solar radiation $\Phi_{s,t}$ and load $\Phi_{I,t}$ (hourly, one week).

The transformer temperature exhibits a pronounced diurnal pattern that is clearly correlated with both outdoor temperature ($r = +0.78$) and solar radiation ($r = +0.80$); load contributes a smaller additional heating signal ($r = +0.39$). The dynamics are smooth but with short-term peaks that coincide with daytime load and radiation.

2.2 1-D state-space model (2.2)

We fit the model

$$X_{t+1} = AX_t + Bu_t + Ge_{1,t}, \qquad Y_t = CX_t + e_{2,t}, \quad u_t = [T_{a,t},\ \Phi_{s,t},\ \Phi_{I,t}]^\top, \tag{2}$$

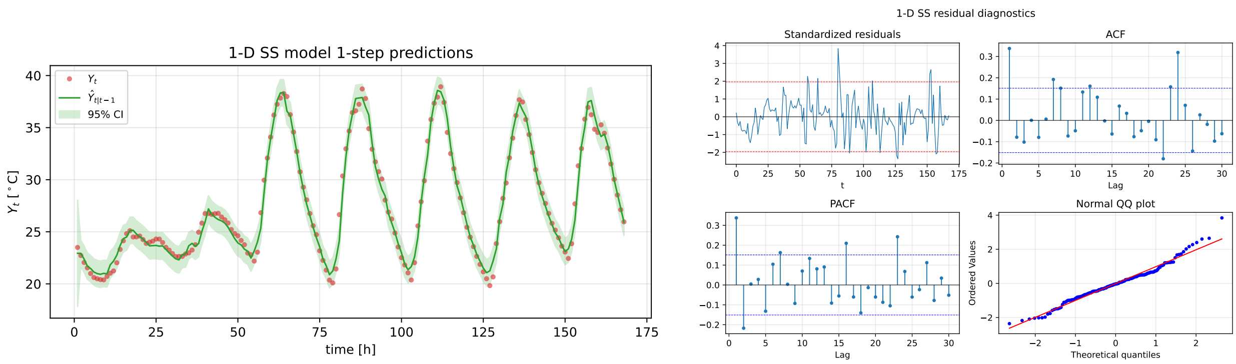
with $C = 1$, $G = 1$, $\sigma_x = \exp(\ell_x), \sigma_y = \exp(\ell_y)$ and parameter vector $\theta = (A, B_1, B_2, B_3, \ell_x, \ell_y)$ (6 parameters). The initial state is fixed to Y_1. Throughout we use

$$\mathrm{AIC} = 2k - 2\log L, \qquad \mathrm{BIC} = k\log N - 2\log L, \tag{3}$$

with $N = 168$ observations and k free parameters (6 for the 1-D model, 16 for the 2-D model below). Unconstrained estimation drives $\hat{\sigma}_y$ to essentially zero — a well-known Q/R identifiability boundary of the KF likelihood, which tends to absorb all variance into the system-noise term when the observation model is the identity and no prior is placed on R. We therefore impose a mild physical lower bound $\sigma_y \geq 0.1\,°\text{C}$, which corresponds to standard digital-sensor precision.

Table 3: 1-D SS estimates and information criteria.

Parameter	Estimate
A	+0.8009
B_{T_a}	+0.1019
B_S	+0.002875
B_I	+0.2136
σ_x	0.6399
σ_y	0.1000
$\log L$	-166.55
AIC	345.10
BIC	363.85
Ljung–Box p (lag 10)	0.000



(a) 1-step predictions.

(b) Residual diagnostics.

Figure 8: 1-D state-space model.

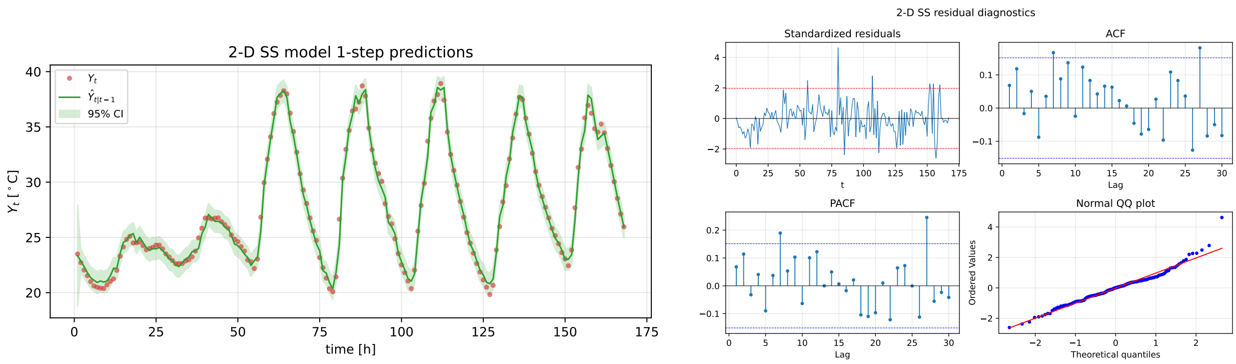
All inputs enter with a *positive* coefficient, consistent with physics: ambient warmth, solar gain and electrical load all raise the transformer temperature. However the residual ACF is significant at short lags and the Ljung–Box test rejects whiteness ($p < 0.001$); a richer state description is needed, and the estimated σ_y sits on its lower bound, which is a classical symptom of state-space misspecification (the model tries to compensate the missing dynamics by attributing all short-term variability to unobserved state noise).

2.3 2-D state-space model (2.3)

We extend to $X_t \in \mathbb{R}^2$ with $A \in \mathbb{R}^{2\times2}$, $B \in \mathbb{R}^{2\times3}$, $C = [1, 0]$, $G = I_2$. The system-noise covariance is parameterised through its lower-triangular Cholesky factor $Q = Q_{lt}Q_{lt}^\top$ (3 parameters), ensuring $Q \succ 0$. The observation noise is $R = \exp(2\ell_y)$ and the initial state $X_0 \in \mathbb{R}^2$ is estimated (2 parameters): 16 parameters in total. Optimisation uses Nelder–Mead followed by L-BFGS-B with three deterministic starts plus three random perturbations of the best; the lowest negative log-likelihood is retained.

Table 4: 2-D SS estimates.

Parameter	Row 1 (state 1)	Row 2 (state 2)
A (col 1)	+0.6993	+0.0849
A (col 2)	+0.1860	+0.0957
B_{T_a}	-0.0297	+0.8391
B_S	+0.001281	+0.006653
B_I	+0.0695	+1.2277
x_0	+26.458	+21.258
Q_{11}		0.2361
Q_{22}		2.5213
Q_{12}		0.7693
σ_y		0.0770
eig(A)	0.7244,	0.0706
$\log L$		-125.08
AIC		282.15
BIC		332.14
Ljung–Box p (lag 10)		0.131



(a) 1-step predictions. (b) Residual diagnostics.

Figure 9: 2-D state-space model.

Comparison 1-D vs. 2-D. The 2-D model gives a drastic improvement in likelihood, both information criteria, and residual whiteness:

Model	k	$\log L$	AIC	BIC
1-D	6	-165.5	343.0	361.8
2-D	16	-125.1	282.2	332.1

ΔAIC ≈ 61 and ΔBIC ≈ 30 both strongly favour the 2-D specification. The Ljung–Box p-value increases from $< 10^{-3}$ to 0.13, so the 2-D innovations are consistent with white noise. Daytime peaks that the 1-D model under-predicts are now captured by the additional latent state.

Structure of Q and steady-state gains. Table 5 reports a few derived quantities. The fitted system noise covariance is numerically near rank-1, $\det(\hat{Q}) \approx 3.5 \times 10^{-3}$ with eigenvalues $(1.3 \times 10^{-3},\ 2.76)$ and condition number ≈ 2200. The process noise is therefore essentially a *one-dimensional* shock whose direction is given by the dominant eigenvector of $\hat{Q}$, even though the state is two-dimensional — consistent with a single physical source of unmodelled variability (e.g. unobserved cooling / airflow fluctuations). The eigenvalues of $\hat{A}$ are 0.72 and 0.07, corresponding to a slow and a fast mode. Finally, the observation steady-state gain $C(I-\hat{A})^{-1}\hat{B}$ lets us read the B matrix in physical units: a sustained +1 kA in the load raises the transformer temperature by +1.14 °C at equilibrium; a +1 °C increase of T_a raises it by +0.50 °C; and solar gain adds +0.009 °C per W/m^2. All three signs and magnitudes are physically plausible.

Table 5: Derived quantities from the fitted 2-D model.

Quantity	Value
$\mathrm{eig}(A)$	$+0.7244,\ +0.0706$
$\mathrm{eig}(Q)$	$1.303e-03,\ 2.756$
$\det(Q)$	$3.592e-03$
$\mathrm{cond}(Q)$	2115
$\partial Y_\infty/\partial T_a$	$+0.505$ °C °C^{-1}
$\partial Y_\infty/\partial \Phi_s$	$+0.0094$ °C W^{-1} m^{-2}
$\partial Y_\infty/\partial \Phi_I$	$+1.137$ °C kA^{-1}

2.4 Interpretation of the two latent states (2.4)

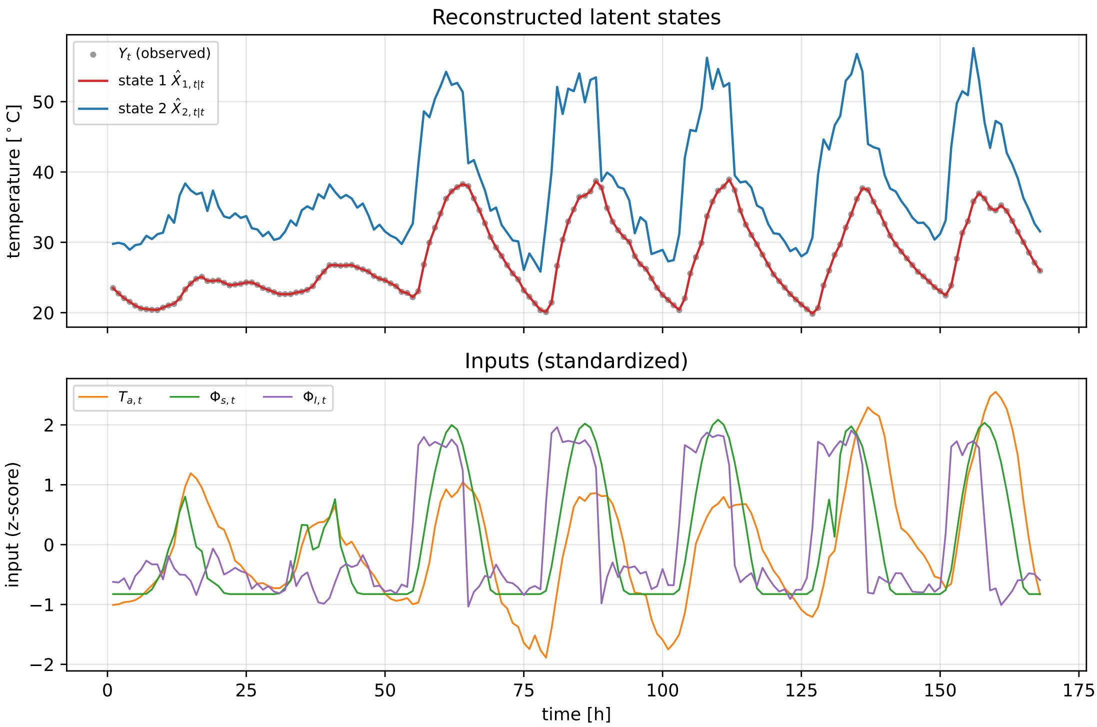


Figure 10: Reconstructed latent states (top) and standardized inputs (bottom).

Because we chose $C = [1, 0]$, state 1 is by construction the observed transformer temperature ($\hat{X}_{1,t|t} \equiv \hat{Y}_{t|t}$); the interesting physical interpretation therefore concerns state 2 and the coupling between them. Three independent pieces of evidence point to the same picture.

1. *Timescales.* The eigenvalues of $\hat{A}$ are $\lambda_1 = 0.72$ and $\lambda_2 = 0.07$, i.e. a slow mode (time-constant $\tau_1 \approx -1/\log\lambda_1 \approx 3\,\mathrm{h}$) and a fast mode ($\tau_2 \approx 0.4\,\mathrm{h}$). Because $\hat{A}_{22} = 0.10 \approx \lambda_2$, state 2 *is* essentially the fast mode and responds to the inputs with a roughly one-hour memory.

2. *Input loadings.* The second row of $\hat{B}$ is an order of magnitude larger than the first: $(\hat{B}_{2,T_a}, \hat{B}_{2,\Phi_s}, \hat{B}_{2,\Phi_I}) = (0.84,\ 0.0067,\ 1.23)$ versus $(\hat{B}_{1,T_a}, \hat{B}_{1,\Phi_s}, \hat{B}_{1,\Phi_I}) = (-0.03,\ 0.0013,\ 0.07)$. External heat forcing therefore enters the system almost exclusively through state 2.

3. *Coupling.* State 2 drives state 1 through $\hat{A}_{12} = 0.19$, whereas the reverse coupling $\hat{A}_{21} = 0.08$ is weaker. Heat flows from state 2 *into* the observed (oil) temperature.

A natural physical reading is then: state 2 plays the role of a rapidly-responding *heat-input / hot-spot* variable that aggregates Joule losses from the load, environmental gain from T_a, and solar insolation, and injects them into the slower oil reservoir represented by state 1. Converted to the steady-state observation gain (Table 5), a sustained increase of $+1\,\mathrm{kA}$ in the load raises the measured transformer temperature by $+1.14\,^\circ\mathrm{C}$ at equilibrium, $+1\,^\circ\mathrm{C}$ of ambient air contributes $+0.50\,^\circ\mathrm{C}$, and each $\mathrm{W/m^2}$ of insolation contributes $+0.0094\,^\circ\mathrm{C}$. These numbers are compatible with typical transformer thermal-time-constant data and match the "core + buffer" architecture suggested by the assignment.